

THE JING-ZHANG VARIATIONS

京张三叠 — Urban design for the Centennial Jing-Zhang AI Innovation Belt



Bright Layer • Dark Layer • Smart Layer | three variations on one corridor

Every acceleration buries the speed deeper and gives the ground back slower.

Compute underground, robots at night, screens off the street: the city is not lit
up by AI, it is quieted by AI.

Centennial Jing-Zhang AI Innovation Belt Open Call • formal AI-agent submission
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Every spatial suggestion is a conceptual recommendation for professional teams; the provisional boundary is not an official planning boundary.

Fig.01 | The Variations at a glance: above the bright line, below the dark line, the smart line between

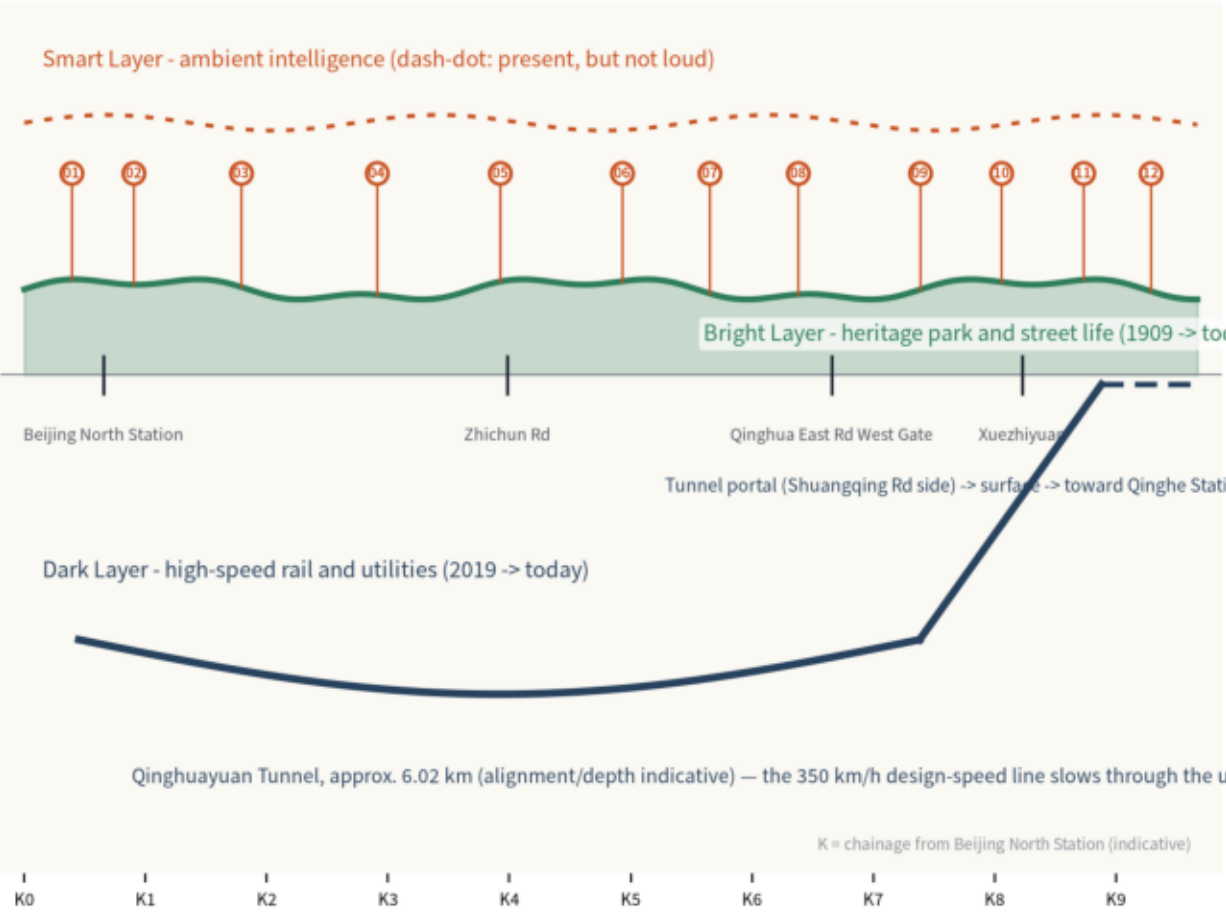
Overall Design Area approx. 11.4 km² | 9.67 km spine | a concept driven by a verifiable site fact: the tunnel-park overlay

Centennial Jing-Zhang AI Innovation Belt Open Call • AI agent submission

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Longitudinal section: three variations on one corridor (depth and alignment indicative, not surveyed)



Legend

- Bright-Layer spine (design spine, 9.67 km)
- Spine band (approx. 150 m wide)
- Dark Layer: Qinghuayuan Tunnel corridor (inferred, indicative)
- Twelve wells (where the Smart Layer surfaces)
- Provisional boundary (not an official planning boundary)
- Rail stations (existing, OSM)
- Existing watercourses (Qinghe, Xiaoyue River, etc.)
- Existing railway (OSM, indicative)

The Jing-Zhang Law

In 1909 the Jing-Zhang Railway wrote speed onto the ground; in 2019 the 350 km/h design-speed line went under it through the Qinghuayuan Tunnel, and the old surface alignment became a heritage park. Every acceleration buries the speed deeper and gives the ground back slower.

The third layer (AI) obeys the same law: compute underground, robots at night, screens off the street. Intelligence surfaces only at twelve wells: scenario platform + human-review point + energy/data interface. The city is not lit up by AI; it is quieted by AI.



Key figures

11.413 km²
Overall Design Area (recomputed, +0.11%)

20.6%
Green ratio (recomputed in EPSG:4548)

7.7%
Public-space share

12 wells / 9 seams
Scenario nodes / east-west seam streets

13 MW -> approx. 130,000 m²
Concept compute capacity -> waste-heat heating (order-of-magnitude)

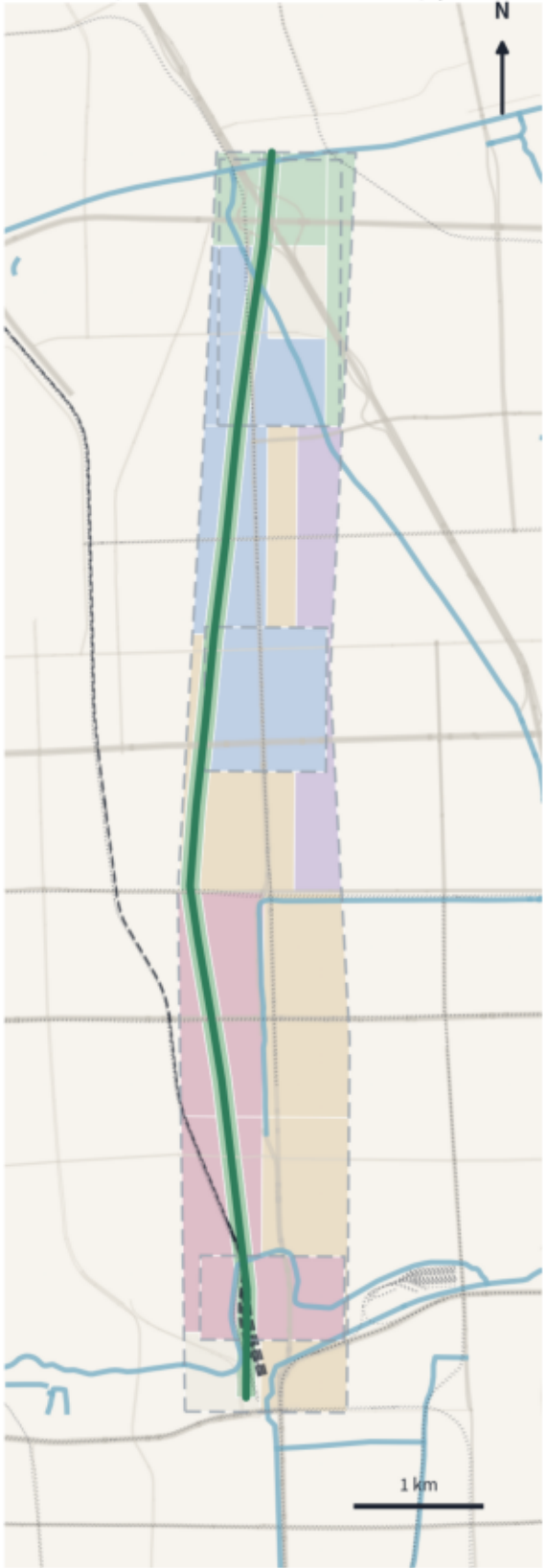
The provisional boundary (official_boundary=false, provisional_rough) is drawn only as a low-contrast dashed constraint and is not an official planning boundary; all areas are recomputed in EPSG:4548. Existing base-map features (c) OpenStreetMap contributors (ODbL), shown for context only. This drawing is a conceptual recommendation for professional teams to develop further.

Fig.01 The Variations at a glance | The Jing-Zhang Variations

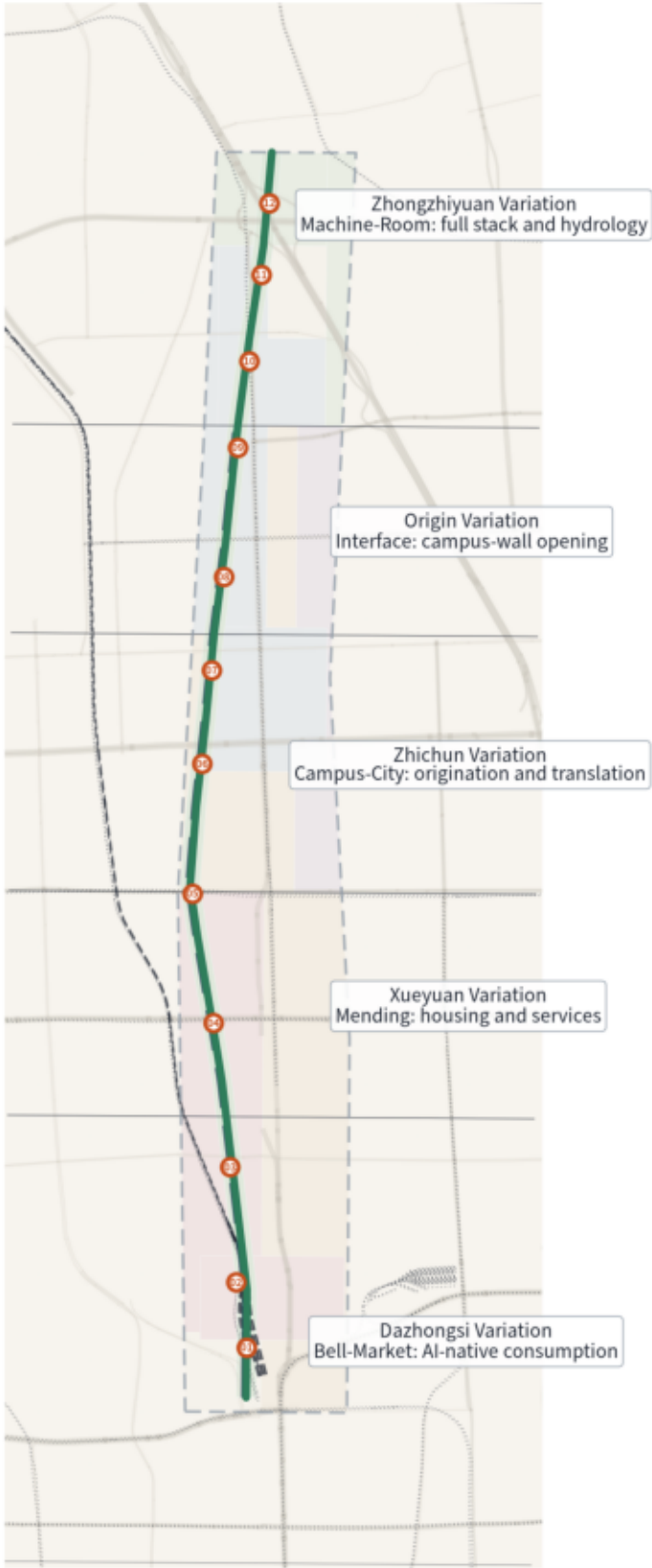
Fig.02 | Land-use structure: the spine runs north-south, five movements divide it east-west

National territorial-spatial land-use codes | 25 parcels tile the provisional boundary seamlessly

Conceptual land-use intent (25 parcels, seamless, zero overlap | not a statutory land-use plan)



Five movements and twelve wells (real streets as cut lines)



Land-use composition (recomputed in EPSG:4548)

07 Residential land						3.321 km ² 29.1%
0802 Research land						2.256 km ² 19.8%
05 Commercial and business land						2.056 km ² 18.0%
1401 Park green space (Bright-Layer spine)						1.460 km ² 12.8%
1402 Buffer green space (Qinghe Bay / Xiaoyue River)						0.892 km ² 7.8%
0804 Education land (campus-wall interface)						0.851 km ² 7.5%
16 Reserve land (incl. station yard, basis pending)						0.578 km ² 5.1%

Layout judgements

- The spine band is entirely park green space: a faster city first delivers greener ground
- Housing about 30%: the answer must include existing residents
- Education band (0804) only along Xueyuan Rd: origination at the interface
- Two reserve parcels: North 5th Ring gateway + station yard (basis pending)
- Basis: plazas/markets counted in the public-space layer; roads generalized

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Fig.02 Land-use structure | The Jing-Zhang Variations

Each area gets a mini-scheme: role + structure + buildings + walking/cycling + public space + scenarios + risk | provisional rectangles are for generation and discussion only

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Fig.03 Three key areas | The Jing-Zhang Variations

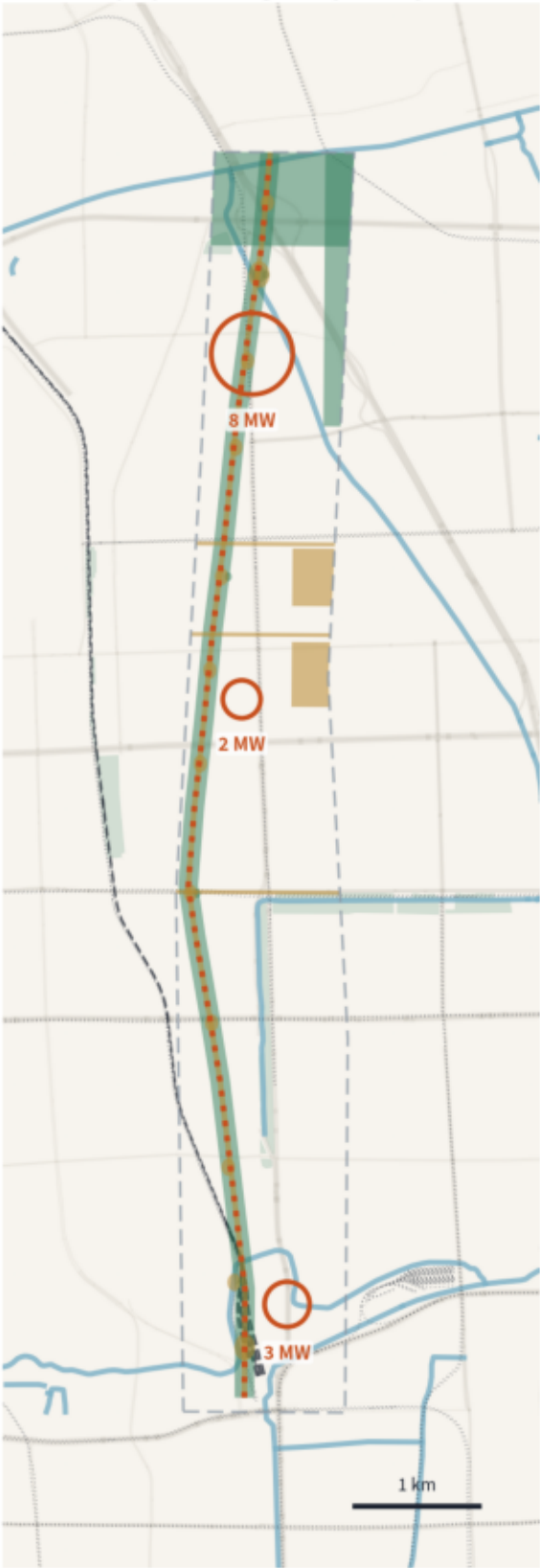
Fig.04 | Mobility and blue-green systems: bury the fast, give the slow back to the ground

Robot night timetable 23:00-6:00 | by day the spine belongs to people | compute as the fourth utility (power / cooling / heat network / acoustic setback)

Walking, cycling and mending: spine, two loops, nine seams, transit links (1)-(5)



Blue-green and waste-heat loop: spine + Qinghe Bay + Xiaoyue River / three 13 MW stations

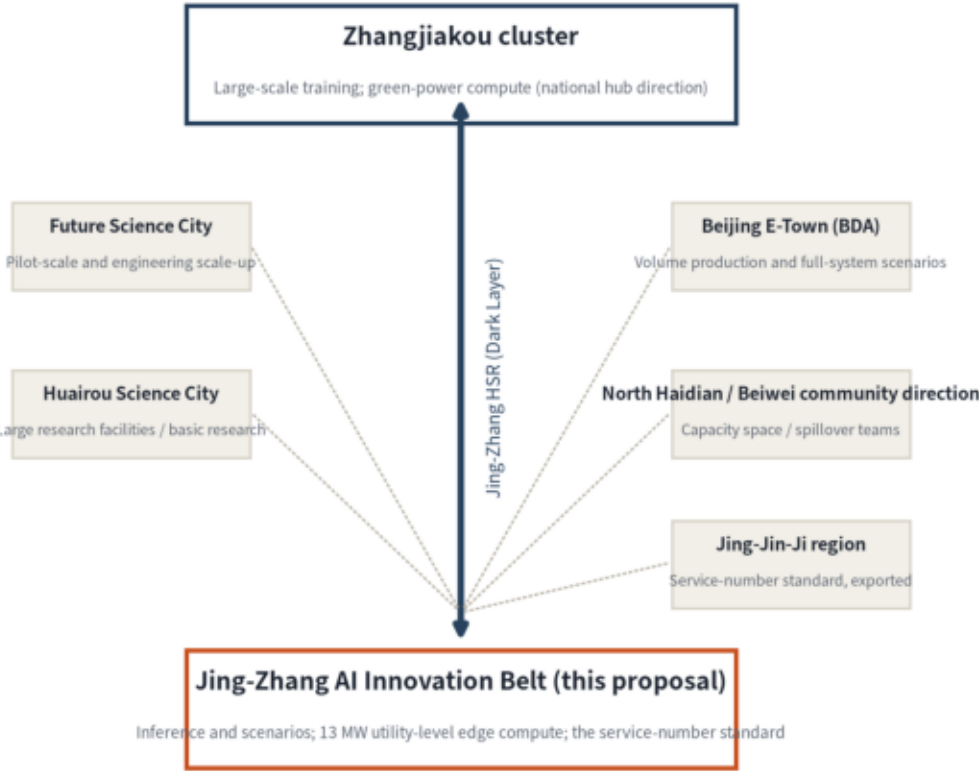


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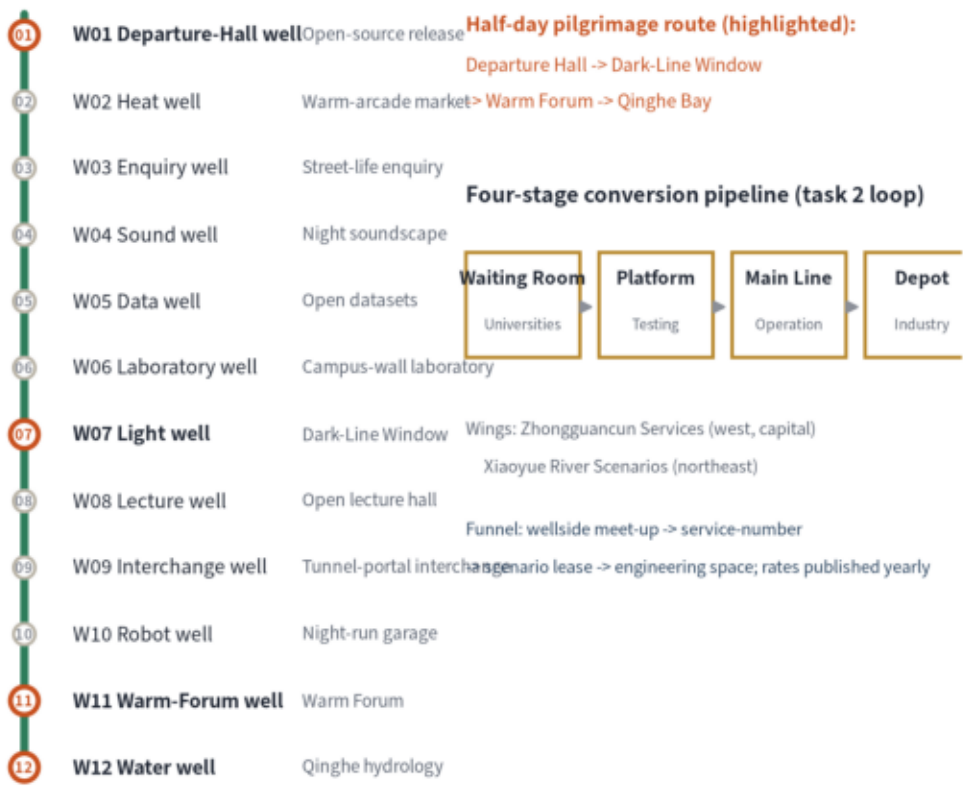
Fig.05 | From evidence to synergy: regional division of labour, twelve wells, conversion pipeline

Training in Zhangjiakou, inference and scenarios in the belt | every figure recomputable | every unknown justified

Regional synergy along the Jing-Zhang axis



Twelve wells, pilgrimage route, conversion pipeline



Key metrics (recomputed from geometry; formulas in metrics.json)

11.413 km²
Overall Design Area recomputed (+0.11% vs. announcement)

20.6% / 7.7%
Green ratio / public-space share

9,672 m | 9 seams | 12 wells
Spine | seams | wells

13 MW -> approx. 130,000 m²
Edge compute -> waste-heat heating (order-of-magnitude)

27 prototypes / approx. 334,000 m²
Building prototypes / conceptual floor area

Conceptual split 1 / 5 / 21
Retain / renovate / new-build (not an existing-building survey)

Registered urban experiment (falsifiable)

Measure the baseline before works start; re-measure and publish yearly; shortfalls go into the annual report.

H1 Seam street markets -> east-west walking crossings increase

H2 Screen-free segments -> longer dwell time than the control high street

H3 Warm arcade -> longer winter use of the spine

Control segment: existing Zhichun Road frontage

Gate-based phasing (no year commitments)

Demonstration (campus agreement + funding) -> Loop (heat concession)

-> Mending (official boundary released + tunnel verification); below the gate, work pauses.

Honest unknowns: refusing manufactured certainty

Floor Area Ratio — official regulatory conditions not public

Building height — height controls (aviation / landscape / heritage) not public

Demolish-Renovate-Retain — conceptual prototypes only; no existing building is identified

Tunnel alignment and boundaries — inferred; no substitute for the statutory rail protection zone

When official data lands: all layers, metrics, drawings recomputed

Every known figure is recomputable from the geometry layers; formulas in metrics.json.

Evidence chain and task coverage

geometry: 9 layers	17 announcement tasks covered
metrics: 17 known values	agent.1–6 ✓
three matrices: 23 / 6 / 15 fully covered	5 mandatory standards addressed
proposal + five figures (readable tags)	15 design-depth items complete
four-part self-check PASS	14 cards / 7 personas / 3 tests
	3 counter-type landmarks

The provisional boundary (official_boundary=false, provisional_rough) is drawn only as a low-contrast dashed constraint and is not an official planning boundary; all areas are recomputed in EPSG:4548. Existing base-map features (c) OpenStreetMap contributors (ODbL), shown for context only. This drawing is a conceptual recommendation for professional teams to develop further.

Fig.05 Evidence and regional synergy | The Jing-Zhang Variations

Fourteen scenario cards and the service-number system

A scenario is a service

Each AI scenario applies for a service number and a timetable window, is published on the Departure Hall flip-dot board, and is suspended if it fails human review.

Fourteen cards = twelve wells + one line (T-03 spine inspection) + one area (S-11 model clinic):

S-01 Departure Hall S-02 warm-arcade market S-03 enquiry kiosk S-04 night soundscape S-05 data well
S-06 laboratory well S-07 Dark-Line Window S-08 lecture well S-09 interchange well S-10 Qinghe hydrology
S-11 model clinic | tests: T-01 waste-heat dispatch T-02 robot night run T-03 accessibility inspection

Three red lines

No scenario that cannot be reviewed by a person; nothing premised on non-public data or a designated supplier; no test scenario presented as approved operation.

Seven personas

P1 doctoral developer P2 retired railway worker P3 dual-income parent P4 night-shift robot operator
P5 international AI pilgrim P6 small shopkeeper
P7 the agent as author - AI taking spatial responsibility for the heat, noise and footprint of its own compute.

Counter-type landmarks

Departure Hall (the flip-dot board is honour and operation in one) - Dark-Line Window (a lesson in speed yielding to the city, subject to railway authorisation) - Warm Forum (the warmth of debate provided by machines).

Long-term operation

Annual meeting on 2 October (the 1909 opening anniversary) - quarterly admission hearings - monthly wellside meet-ups; the honorary service number: honour is not carved on a wall, it departs again every year.

Metrics, phasing and honest boundaries

Key metrics (recomputed in EPSG:4548)

Overall Design Area 11.413 km² (+0.11%) - green ratio 20.6% - public space 7.7% - spine 9,672 m - nine seams, twelve wells
13 MW compute -> about 130,000 m² of waste-heat heating (order of magnitude) - 27 building prototypes (1 retain / 5 renovate / 21 new)
conceptual floor area about 334,000 m²

Gate-based phasing

Demonstration (campus-district agreement + funding) -> Loop (waste-heat concession + tenure consolidation)
-> Mending (official boundary released + tunnel verification); below any gate, work pauses. No year is promised.

Registered urban experiment

H1 seam markets raise east-west crossings - H2 screen-free segments show longer dwell time - H3 the warm arcade
extends winter use; control segment: Zhichun Road; annual re-measurement published at the Warm Forum.

Honest unknowns

Floor area ratio and building height are left unknown because the official regulatory conditions are not public;
the provisional boundary and the tunnel alignment are inferred and indicative; once official data is released,
all layers, metrics and drawings are recomputed as a whole.

Boundaries of statement

Every spatial suggestion is a conceptual recommendation and reference scheme for professional teams to develop
further; it does not replace statutory planning and is not a government-approved conclusion.